Tensor Decomposition-Based Relaxed Linear Regression for Hyperspectral Image Classification

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***Abstract*— Linear regression and its variants have achieved considerable success in image classification. However, those methods still encounter two challenges when dealing with hyperspectral image (HSI) classification. On the one hand, the existing ones focus on mining the relationship between the label space and original data space during the classifier training, which is generally sensitive to noise corruptions. On the other hand, transforming the training samples into a strict binary label matrix makes the generalization ability of the classifier limited. To address these challenges, this article constructs a novel integrative model called tensor decomposition-based relaxed linear regression (TDRLR) for HSI classification. First, the model adopts tensor canonical polyadic (CP) decomposition to learn two dictionaries from spatial and spectral directions, respectively, which can help to generate a double dictionary representation for HSI data. Then, the linear regression classifier is integrated to learn a transformation that reveals the mapping relation between the double dictionary representation and label space rather than the original data for enhancing robustness. Meanwhile, a more flexible way, label relaxation (LR), is employed to enlarge the margins between different classes. More importantly, the learned double dictionary representation and classifier can be fine-tuned in tandem to enhance performance through the designed alternate iterative joint learning algorithm. Experiments conducted on four real-world HSI datasets demonstrate that the proposed method achieves significant improvements in classification performance with a small size training set, when compared with state-of-the-art HSI classification methods.**

***Index Terms*— Classification, hyperspectral image (HSI), relaxed linear regression, tensor decomposition.**

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I. Introduction

RECENT advances in sensor technology have greatly improved the spectral and spatial resolution of remote sensing images. Hyperspectral images (HSIs), which contain a wealth of spectral and spatial information, have proven valuable in studying surface features [1], [2] and supporting various applications in Earth science [3], [4], mineral exploration [5], [6], [7], environmental monitoring [8], and smart agriculture [9]. Despite these benefits, the ultrahigh resolution and high-order data structure of HSIs present significant challenges for classification. Chief among these are the "curse of dimensionality" and the need for effective ways to integrate spatial and spectral information. Furthermore, labeling HSIs can be time-consuming and expensive, often resulting in limited training data. Consequently, there is a growing need for efficient methods that can handle high-order, high-dimensional HSI data and achieve accurate classification with minimal labeled samples.

Numerous methods have been developed for HSI classification, which can broadly be categorized into two main types: divide-and-conquer methods and "end-to-end" integrated network methods. Divide-and-conquer methods achieve HSI classification in two distinct steps, i.e., representation learning and classification. Those methods attempt to learn a robust representation of HSI data using techniques such as feature extraction, feature selection, or dictionary-based representation. The goal is to capture the most meaningful features of the data that can effectively distinguish between different classes. Once suitable data representation is achieved, an existing classifier can be applied to perform the classification task. The primary focus here is on optimizing the data representation. In the early stages of HSI research, classical feature extraction methods like principal component analysis (PCA) [10], linear discriminant analysis (LDA) [11], locality preserving projections [12], and manifold learning (ML) [13] were successfully applied for HSI representation learning. The introduction of the graph embedding (GE) [14] framework further advanced the application of ML in HSI representation. GE-based methods model the data's intrinsic structure by embedding it into a lower dimensional manifold that preserves its essential characteristics. Building on this idea, various graph-based HSI representation learning methods

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have emerged, including sparse graph-based discriminant analysis (SGDA) [15], sparse and low-rank graph-based discriminant analysis (SLGDA) [16], and kernel SLGDA (KGDA) [17].

To effectively leverage the inherent spatial information and accommodate their natural 3-D data structure, tensor theory, which can directly handle multidimensional data, has been introduced into HSI processing. This approach has led to the development of various tensor subspace learning methods under the GE framework, significantly advancing HSI classification. The tensor locality preserving projection (TLPP) method [18] was one of the pioneering methods that applied tensor theory to HSI feature extraction. TLPP was shown to significantly improve classification accuracy by preserving the local neighborhood structure of the data in its tensor form, outperforming traditional vector-based methods. Thus, tensor-based methods, such as modified TLPP (MTLPP) [19], tensor SLGDA (SLGDA) [20], patch tensor-based sparse and low-rank graph (PT-SLG) [21], and GE in core tensor space (GECS) [22], were successively proposed for HSI data representation. After feature extraction using tensor subspace learning methods, a simple classifier can be employed for classification, such as k-nearest neighbors (KNNs) [23], linear regression [24], support vector machines [25], random forests [26], and extreme learning machines [27]. Although tensor-based methods have shown significant advancements in HSI classification, they typically involve a stepwise process where feature extraction and classification are conducted separately. Each step requires optimization of an objective function, which may lead to inefficiencies or the potential of error accumulation across stages.

Due to the powerful feature learning capabilities of deep network structures, the "end-to-end" integrated network models have emerged as powerful tools in various fields of pattern recognition, including HSI processing. These models stand out for their ability to directly learn from raw data, integrating feature extraction and classification into a single process. The typical methods include deep belief network (DBN) [28], convolutional neural network (CNN) [29], [30], [31], CapsuleNet [32], and LSTM [33], which have led to significant improvements in classification performance. To fully exploit the rich spatial information in HSIs, attention mechanisms designed to capture relevant local spatial information have been integrated into deep learning models. A variety of new deep learning suitable for HSI classification such as residual space-spectrum attention network [34] and dense CNN [35] based on feedback attention. A common challenge for these methods is the scarcity of labeled samples, as collecting and annotating these samples can be both costly and time-consuming. Advanced techniques such as generative adversarial networks (GANs) [36], active learning [37], and transfer learning [38] have been developed to address this challenge.

Since the graph convolutional networks (GCNs) [39] have shown great effectiveness in a non-Euclidean space, they have been widely applied for classification by treating HSI as a single graph or a set of graphs [40], [41]. For instance, Qin et al. [42] proposed a spatial–spectral graph convolutional networks (S^2GCNs), utilizing adjacency nodes in the graph to approximate convolution. Ding et al. [43] proposed the global consistent graph convolutional network (GCGCN), which constructs a reliable local graph at a global level to capture rich spatial information by maintaining topological consistency. In order to extract nonlinear spatial–spectral features, Wang et al. developed the broad graph convolutional neural network [44], integrating a broad learning system with GCNs to enhance HSI feature extraction through graph convolution operations. From a superpixel perspective, Jia et al. proposed the graph-in-graph (GiG) model and its corresponding GiG convolutional network (GiGCN) for HSI classification [45]. Dong et al. [46] combined superpixel-based graph attention networks (GAT) with pixel-based CNN to capture complementary features, thereby improving classification performance. Besides the aforementioned methods, transformer-based methods have been widely applied in HSI classification. For instance, Ouyang et al. [47] developed a hybrid transformer network that integrates spatial–spectral attention and multigranularity tokens for feature extraction. Shi et al. [48] introduced a dual-branch multiscale transformer network, which combines spectral convolution kernel and channel shrink soft split module for multiscale spatial–spectral feature. While these deep learning methods have significantly advanced HSI classification, they generally require a large number of well-labeled training samples. Moreover, the computational complexity of these deep learning models is often high, imposing difficulty in their practical application.

Inspired by the "end-to-end" integrated network models, the manner of integrating feature extraction (or representation learning) and classifier training together in one model, has gained considerable attention due to its potential to enhance data processing efficiency. In this context, many novel approaches have been proposed. Yang et al. introduced orthogonal autoencoder regression (OAR) [49], which integrates feature learning with regression representation to create a linear autoencoder for feature extraction and classification. A novel method called $L_{2,p}$-norm-based robust embedding regression ($L_{2,p}$-RER) [50] was proposed, which combines projection learning with robust PCA (RPCA) to effectively suppress noise and reduce feature dimensions. Cheng et al. [51] proposed the reweighted robust and discriminative latent subspace projection (ReRDLSP) method, which jointly minimizes subspace representation and relaxed ridge regression. Bao et al. [52] constructed an integrated model named robust embedding regression (RER), leveraging existing robust linear regression and low-rank representation (LRR) techniques to reduce redundant features and suppress noise. However, those methods fail to effectively capture the spatial context information for representation learning when dealing with HSIs, which is crucial for representation learning in this domain.

To address the above issue, this article constructs a novel integrative model called tensor decomposition-based relaxed linear regression (TDRLR) for HSI classification, as illustrated in Fig. 1. During the model training and testing, it constructs new spectral–spatial tensor representations of training and

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testing sets by unfolding the local patch-based tensors of HSI. At the training stage, the proposed model contains two modules, i.e., tensor decomposition-based double dictionary representation and relaxed linear regression classifier. Specifically, tensor canonical polyadic (CP) decomposition is first utilized to decompose the spatial–spectral tensor representation of the training set for learning two shared spatial and spectral feature dictionaries. Simultaneously, spectral and spatial dictionaries collaboratively represented double dictionary feature representation of the training set is yielded for classifier training. Then, a discriminative transformation matrix of the classifier is learned for classification by exploiting the correlation between the double dictionary feature representation and label space. In particular, the label relaxation (LR) trick and $L_{2,1}$-norm constraint are employed to ensure the discriminability and robustness of the classifier. At the testing stage, the double dictionary feature representation of the testing set can be first generated by tensor CP decomposition with the learned common spectral and spatial feature dictionaries. Then, the predictive labels of the testing set can be obtained by utilizing the well-trained relaxed linear regression classifier to directly classify the double dictionary representation of the testing set. In summary, the main contributions of this article are as follows.

1) TDRLR, an integrative model that can achieve the joint learning of spatial–spectral double dictionary representation and relaxed linear regression classifier, is the first time proposed and applied in the field of HSI classification.
2) Different from previous methods, TDRLR utilizes tensor CP decomposition to learn a spatial dictionary and a spectral dictionary for collaboratively encoding both spatial and spectral structural information of HSIs. Meanwhile, LR is embedded into the linear regression regularization to enlarge interclass distance and $L_{2,1}$-norm constraint is imposed to improve the robustness of the classifier.
3) Moreover, an iterative jointly learning algorithm is designed to optimize the proposed model, which ensures that the multilinear spatial–spectral structure of HSI can be explored to enhance the robustness of the classifier and the prior label information can be embedded to improve the discriminability of the double dictionary representation during the model training.
4) Extensive experiments are conducted on four public HSI datasets to verify the effectiveness of the proposed TDRLR method. The associated experimental results demonstrated that TDRLR can achieve more excellent classification performance on four datasets than some state-of-the-art methods when using small sizes of training sets.

This article is organized as follows. Section II reviews related work. Section III presents the motivation behind the proposed TDRLR model, details the solution algorithm, and discusses its computational complexity. Section IV provides experimental results obtained from four datasets and includes an ablation analysis. Finally, the conclusion is drawn in Section V.

II. Related Work

A. Tensor Fundamentals

A tensor is a multidimensional array [18]. The order of a tensor refers to the number of dimensions, also known as ways or modes. More formally, an Nth-order or N way tensor is an element of the tensor product of N vector spaces. Throughout this article, lowercase letters (e.g., a), bold capital letters (e.g., $\mathbf{a}$), bold lowercase letters (e.g., $\mathbf{A}$), and calligraphic letters (e.g., $\mathcal{A}$) are used for scales, vectors, matrices, and tensors, respectively. $\mathcal{A}_{(:,:,i)}$ denotes the ith frontal slice matrix of tensor $\mathcal{A}$. The definitions [53] related to this work are given as follows.

Definition 1 (k-Mode Unfolding Form): k-mode unfolding expands a tensor into a matrix. Let $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$ be a tensor of order N, the k-mode unfolding of $\mathcal{X}$ is denoted as $\mathcal{X}_{(k)} \in \mathbb{R}^{I_k \times (I_N I_{N-1} \cdots I_{k+1} I_{k-1} \cdots I_1)}$, which arranges the mode-k fibers to be the columns of the matrix.

Definition 2 (Rank-One Tensors): An N-mode tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$ is rank one if it can be written as the outer product of N vectors, i.e.,

$$\mathcal{X} = \mathbf{a}^{(1)} \circ \mathbf{a}^{(2)} \circ \mathbf{a}^{(3)} \circ \cdots \circ \mathbf{a}^{(N)} \tag{1}$$

where the symbol "$\circ$" represents the vector outer product.

Definition 3 (CP Decomposition): The CP decomposition factorizes a tensor into a sum of component rank-one tensors. More specifically, given a third-order tensor $\mathcal{X} \in \mathbb{R}^{l \times m \times n}$, its CP decomposition can be written as

$$\mathcal{X} \approx \sum_{r=1}^{R} \mathbf{a}_r \circ \mathbf{b}_r \circ \mathbf{c}_r = [[\mathbf{A}; \mathbf{B}; \mathbf{C}]] \tag{2}$$

where R is a positive integer, and $\mathbf{a}_r \in \mathbb{R}^l$, $\mathbf{b}_r \in \mathbb{R}^m$, $\mathbf{c}_r \in \mathbb{R}^n$, $\mathbf{A} = [\mathbf{a}_1, \ldots, \mathbf{a}_R]$, $\mathbf{B} = [\mathbf{b}_1, \ldots, \mathbf{b}_R]$, $\mathbf{C} = [\mathbf{c}_1, \ldots, \mathbf{c}_R]$. $[[\mathbf{A}; \mathbf{B}]]$ denotes the outer product of matrices $\mathbf{A}$ and $\mathbf{B}$. Elementwise,

$$x_{ijk} \approx \sum_{r=1}^{R} a_{ir} b_{jr} c_{kr} \tag{3}$$

for $i = 1, \ldots, l$, $j = 1, \ldots, m$, $k = 1, \ldots, n$.

Definition 4 (Khatri–Rao Product): Given two matrices $\mathbf{A} \in \mathbb{R}^{I \times N}$ and $\mathbf{B} \in \mathbb{R}^{J \times K}$, their Kronecker product is defined as

$$\mathbf{A} \otimes \mathbf{B} = \begin{bmatrix} a_{11}\mathbf{B} & a_{12}\mathbf{B} & \cdots & a_{1R}\mathbf{B} \\ a_{21}\mathbf{B} & a_{22}\mathbf{B} & \cdots & a_{2R}\mathbf{B} \\ \vdots & \vdots & \ddots & \vdots \\ a_{I1}\mathbf{B} & a_{I1}\mathbf{B} & \cdots & a_{IR}\mathbf{B} \end{bmatrix} \tag{4}$$

which is an $IJ \times NK$ matrix. The Khatri–Rao product is usually defined as the column-wise Kronecker product, which can be expressed as

$$\mathbf{A} \odot \mathbf{B} = [\mathbf{a_1} \otimes \mathbf{b_1} \quad \mathbf{a_2} \otimes \mathbf{b_2} \quad \cdots \quad \mathbf{a_R} \otimes \mathbf{b_R}]. \tag{5}$$

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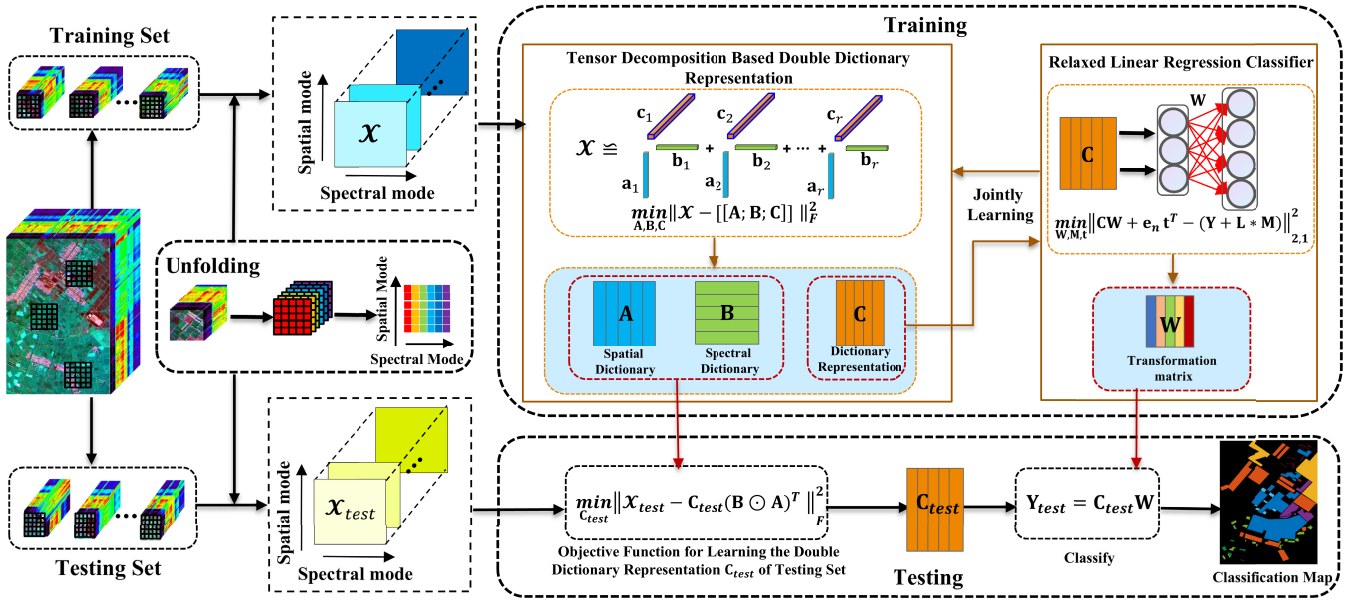


Fig. 1. Flowchart of the proposed TDRLR model for HSI classification. The model includes two modules: tensor CP decomposition-based double dictionary learning and LR for linear regression classifier training.

B. Label Relaxation

The linear regression model can reveal the relationship between sample features and labels through a transformation matrix. However, this model requires converting labels into strict binary matrices, reducing the flexibility of constructing a transformation matrix. To address this limitation, the LR trick introduces a nonnegative slack matrix to transform the strict binary label matrix into a more flexible relaxed label matrix [54]. For example, let

$$\mathbf{Y}_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix} \tag{6}$$

be the original label matrix for four samples, which belong to the first, third, second, and second class, respectively. LR aims to relax the strict binary constraints in $\mathbf{Y}_1$, allowing for more flexible fitting of the labels. Consequently, a relaxed label matrix $\mathbf{Y}_1'$ is used to replace the original label matrix $\mathbf{Y}_1$, which is defined as follows:

$$\mathbf{Y}_1' = \begin{bmatrix} 1+m_{11} & -m_{12} & -m_{13} \\ -m_{21} & -m_{22} & 1+m_{23} \\ -m_{31} & 1+m_{32} & -m_{33} \\ -m_{41} & 1+m_{42} & -m_{43} \end{bmatrix}, \quad \text{s.t. } m_{ij} \geq 0. \tag{7}$$

It can be shown that by setting $m_{ij} \geq 0$, the distance between different classes can be increased, making the transformation matrix more discriminative.

Without loss of generality, suppose $\mathbf{Y} \in \mathbb{R}^{n \times c}$ is an original label matrix of data, and then the relaxed label $\mathbf{Y}'$ can be constructed as

$$\mathbf{Y}' = \mathbf{Y} + \mathbf{L} * \mathbf{M} \tag{8}$$

where $*$ denotes the Hardamard product of the matrix, $\mathbf{L} \in \mathbb{R}^{n \times c}$ is a luxury matrix defined as:

$$\mathbf{L}_{ij} = \begin{cases} +1, & \text{if } \mathbf{Y}_{ij} = 1 \\ -1, & \text{if } \mathbf{Y}_{ij} = 0 \end{cases} \tag{9}$$

and $\mathbf{M}$ is a nonnegative slack matrix devised as

$$\mathbf{M} = \begin{bmatrix} m_{11} & \cdots & m_{1c} \\ \vdots & \ddots & \vdots \\ m_{n1} & \cdots & m_{nc} \end{bmatrix}, \quad \text{s.t. } m_{ij} \geq 0. \tag{10}$$

III. Methodology

A. Model Formulation

As previously mentioned, making full use of both spatial and spectral information of HSIs to accurately express HSIs is a hot topic that has always been widely concerned in the field of HSI processing. Dictionary learning is an effective way to gain low-dimensional representation for high-dimensional data, which is also widely applied for HSI representation. However, the current single dictionary representation methods failed to simultaneously integrate both spatial and spectral information to encode the HSI data. To address this issue, we propose to learn two dictionaries from both spatial and spectral dimensions for collaboratively encoding the samples of HSI with both intrinsic spatial and spectral information well captured. In detail, the local patch-based tensor representations of HSI are first unfolded to construct a new spatial–spectral tensor representation $\mathcal{X} \in \mathbb{R}^{m \times l \times n}$. According to matrix factorization theory, each sample $\mathcal{X}_{(:,:,i)} \in \mathbb{R}^{m \times l}$ can be decomposed into (as illustrated in Fig. 2)

$$\mathcal{X}_{(:,:,i)} = \mathbf{A}\mathbf{C}^i\mathbf{B}^T = \sum_{j=1}^{r} c_j^i \left(\mathbf{a}_j \circ \mathbf{b}_j^T\right) \tag{11}$$

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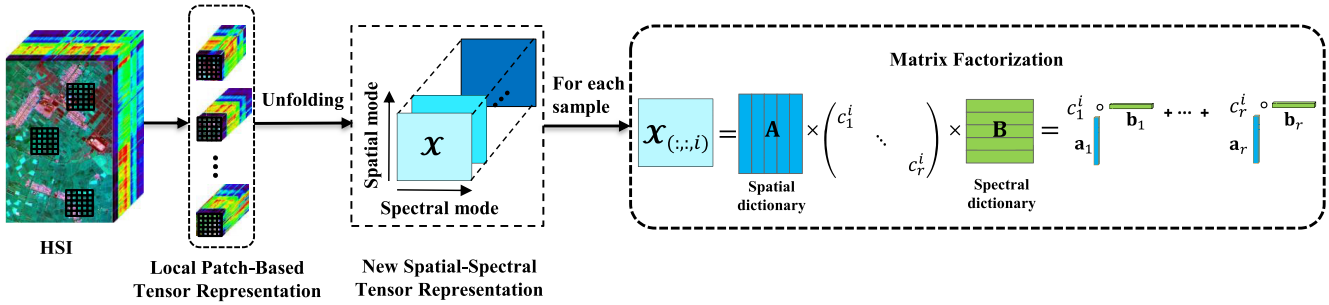


Fig. 2. Illustration of the mechanism for the double dictionary representation of each sample of HSI based on both spatial and spectral dictionaries.

where r is the rank of matrix $\mathcal{X}_{(:,:,i)}$, $\mathbf{A} = [\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_r]$ and $\mathbf{B} = [\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_r]$ can be regarded as two dictionaries from the spatial and spectral directions, respectively. $\mathbf{C}^i = \text{diag}(c_1^i, c_2^i, \ldots, c_r^i)$ is the corresponding coefficient of double dictionary representation. Thus, to learn two shared spatial and spectral feature dictionaries for collaboratively representing all samples, tensor CP decomposition is introduced to formulate a spatial–spectral double dictionary representation learning model. Mathematically, it can be expressed as

$$\min_{\mathbf{A},\mathbf{B},\mathbf{C}} \|\mathcal{X} - [[\mathbf{A}; \mathbf{B}; \mathbf{C}]]\|_F^2 \tag{12}$$

where $\mathcal{X} \in \mathbb{R}^{m \times l \times n}$ is the training set with n tensor objects of order 2, $\mathbf{A} \in \mathbb{R}^{m \times r}$ and $\mathbf{B} \in \mathbb{R}^{l \times r}$ are two dictionaries from the spatial and spectral directions, respectively. $\mathbf{C} \in \mathbb{R}^{n \times r}$ is the tensor subspace representation coefficient matrix, which is also a double dictionary representation that simultaneously encodes both spatial and spectral information of HSI data.

In general, prior label information plays a crucial role in extracting HSI features. However, the absence of label information in the model (12) renders it an unsupervised learning method, which weakens the discriminability of the extracted subspace representation, especially when training samples are very small. To enhance the discriminability of the learned subspace representation, a linear regression regularization term that aims at revealing the double dictionary representation and label space to search a transformation for classification is integrated into the spatial–spectral double dictionary representation learning model as a classifier. Then, the objective function can be expressed as

$$\begin{aligned} \min_{\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{W}} & \|\mathcal{X} - [[\mathbf{A}; \mathbf{B}; \mathbf{C}]]\|_F^2 + \lambda \|\mathbf{CW} - \mathbf{Y}\|_F^2 \\ \text{s.t. } & \mathbf{W}^T\mathbf{W} = \mathbf{I} \end{aligned} \tag{13}$$

where $\mathbf{Y} \in \mathbb{R}^{n \times c}$ is a label matrix. $\mathbf{W} \in \mathbb{R}^{r \times c}$ is a transformation matrix, which plays a key role by transforming the double dictionary representation into the label space, thereby necessitating the learning of discriminative features. However, the degree of freedom of $\mathbf{W}$ is limited, which restricts the model's generalization ability. To enhance the transformation's discriminative and generalization ability, the LR trick [49], [54], [55] is introduced into the regression regularization term to force the regression targets of different classes far away from each other. Thus, the model (13) can be reformulated as

$$\begin{aligned} \min_{\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{W},\mathbf{t},\mathbf{M}} & \|\mathcal{X} - [[\mathbf{A}; \mathbf{B}; \mathbf{C}]]\|_F^2 \\ & + \lambda \left\|\mathbf{CW} + \mathbf{e}_n\mathbf{t}^T - (\mathbf{Y} + \mathbf{L} * \mathbf{M})\right\|_F^2 \\ \text{s.t. } & \mathbf{W}^T\mathbf{W} = \mathbf{I}, \mathbf{M} > 0. \end{aligned} \tag{14}$$

where $\mathbf{e}_n \in \mathbb{R}^n$ is a vector of all 1, $\mathbf{t}^T \in \mathbb{R}^c$ is a translation vector. Unfortunately, many investigations [49], [50], [51] indicate that the Frobenius norm is sensitive to outliers and performs poorly in noisy data, while $L_{2,1}$ norm is more robust to outliers. The $L_{2,1}$ norm encourages row sparsity of the matrix, making it superior in feature selection and subspace learning. Thus, the $L_{2,1}$ norm is employed to replace the Frobenius norm in (14), resulting in the exact TDRLR model, i.e.,

$$\begin{aligned} \min_{\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{W},\mathbf{t},\mathbf{M}} & \|\mathcal{X} - [[\mathbf{A}; \mathbf{B}; \mathbf{C}]]\|_F^2 \\ & + \lambda \left\|\mathbf{CW} + \mathbf{e}_n\mathbf{t}^T - (\mathbf{Y} + \mathbf{L} * \mathbf{M})\right\|_{2,1} \\ \text{s.t. } & \mathbf{W}^T\mathbf{W} = \mathbf{I}, \quad \mathbf{M} > 0. \end{aligned} \tag{15}$$

This model integrates spatial–spectral double dictionary representation learning and linear regression classifier together, which can effectively capture spatial–spectral information of HSI for classification. More importantly, the embedding of label information in the model significantly enhances the discriminability of the double dictionary representation. LR in the model can increase the distance between different classes, giving the transformation matrix $\mathbf{W}$ a higher degree of freedom and further improving the generalization ability of the classifier. Moreover, the $L_{2,1}$ norm constrained on the regression loss can further enhance the robustness against noise and outliers.

B. Solving Algorithm

It is challenging to solve the proposed TDRLR model analytically due to the nonconvex orthogonality constraint and the inclusion of six variables in the model. In the following, we develop an ADMM to address it. To simplify the model (15), an auxiliary variable $\mathbf{E} \in \mathbb{R}^{n \times c}$ is introduced. Model (15) can then be transformed as follows:

$$\begin{aligned} \min_{\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{W},\mathbf{t},\mathbf{M},\mathbf{E}} & \|\mathcal{X} - [[\mathbf{A}; \mathbf{B}; \mathbf{C}]]\|_F^2 + \lambda \|\mathbf{E}\|_{2,1} \\ \text{s.t. } & \mathbf{E} = \mathbf{CW} + \mathbf{e}_n\mathbf{t}^T - (\mathbf{Y} + \mathbf{L} * \mathbf{M}) \\ & \mathbf{W}^T\mathbf{W} = \mathbf{I}, \quad \mathbf{M} > 0. \end{aligned} \tag{16}$$

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Let $\mu > 0$ be the penalty parameter. The augmented Lagrangian function of (16) is defined by

$$\begin{aligned}&\mathcal{L}_\mu(\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{W},\mathbf{t},\mathbf{M},\mathbf{E};\mathbf{P})\\&\quad=\|\mathcal{X}-[[\mathbf{A};\mathbf{B};\mathbf{C}]]\|_F^2+\lambda\|\mathbf{E}\|_{2,1}\\&\qquad+\frac{\mu}{2}\left\|\mathbf{E}-\mathbf{CW}-\mathbf{e}_n\mathbf{t}^T+\mathbf{Y}+\mathbf{L}*\mathbf{M}+\frac{\mathbf{P}}{\mu}\right\|_F^2-\frac{1}{2\mu}\|\mathbf{P}\|_F^2\\&\text{s.t. }\mathbf{W}^T\mathbf{W}=\mathbf{I},\quad\mathbf{M}>0.\end{aligned}\tag{17}$$

where $\mathbf{P}\in\mathbb{R}^{n\times c}$ is a Lagrangian multiplier. The classic ADMM alternates minimization of the augmented Lagrangian function (17) with respect to each variable. The details are addressed as follows.

1) *Updating* $\mathbf{A}$ *and* $\mathbf{B}$*:* When all variables are fixed except for $\mathbf{A}$, and tensor $\mathcal{X}$ takes the 1-mode unfolding form, the optimization function (17) with respect to $\mathbf{A}$ can be expressed as

$$\mathbf{A}^\star=\underset{\mathbf{A}}{\operatorname{argmin}}\left\|\mathcal{X}_{(1)}-\mathbf{A}(\mathbf{C}\odot\mathbf{B})^T\right\|_F^2.\tag{18}$$

Taking the partial derivative of $\mathbf{A}$ and setting it to zero results in

$$\mathbf{A}(\mathbf{C}\odot\mathbf{B})^T(\mathbf{C}\odot\mathbf{B})-\mathcal{X}_{(1)}(\mathbf{C}\odot\mathbf{B})=0\tag{19}$$

where $\odot$ denotes the Khatri–Rao product. Then, $\mathbf{A}^\star$ can be updated by

$$\mathbf{A}^\star=\mathcal{X}_{(1)}\left[(\mathbf{C}\odot\mathbf{B})^T\right]^+\tag{20}$$

where $(\cdot)^+$ denotes the pseudoinverse of $(\cdot)$. Similarly, the optimization function (17) with respect to $\mathbf{B}$ can be expressed as

$$\mathbf{B}^\star=\underset{\mathbf{B}}{\operatorname{argmin}}\left\|\mathcal{X}_{(2)}-\mathbf{B}(\mathbf{C}\odot\mathbf{A})^T\right\|_F^2.\tag{21}$$

Taking the partial derivative of $\mathbf{B}$ and setting it to zero yields

$$\mathbf{B}(\mathbf{A}\odot\mathbf{C})^T(\mathbf{A}\odot\mathbf{C})-\mathcal{X}_{(2)}(\mathbf{A}\odot\mathbf{C})=0.\tag{22}$$

Then, $\mathbf{B}^\star$ can be updated by

$$\mathbf{B}^\star=\mathcal{X}_{(2)}\left[(\mathbf{A}\odot\mathbf{C})^T\right]^+.\tag{23}$$

By using $(x\odot y)^+=((x^Tx)*(y^Ty))^+(x\odot y)^T$, $\mathbf{A}^\star$, $\mathbf{B}^\star$ can be rewritten as

$$\begin{aligned}\mathbf{A}^\star&=\mathcal{X}_{(1)}(\mathbf{C}\odot\mathbf{B})\left(\left(\mathbf{C}^T\mathbf{C}\right)*\left(\mathbf{B}^T\mathbf{B}\right)\right)^+\\\mathbf{B}^\star&=\mathcal{X}_{(2)}(\mathbf{A}\odot\mathbf{C})\left(\left(\mathbf{A}^T\mathbf{A}\right)*\left(\mathbf{C}^T\mathbf{C}\right)\right)^+\end{aligned}\tag{24}$$

respectively.

2) *Updating* $\mathbf{C}$*:* When all variables are fixed except $\mathbf{C}$, and tensor $\mathcal{X}$ takes the three-mode unfolding form, the optimization function (17) with respect to $\mathbf{C}$ can be expressed as

$$\mathbf{C}^\star=\underset{\mathbf{C}}{\operatorname{argmin}}\left\|\mathcal{X}_{(3)}-\mathbf{C}(\mathbf{B}\odot\mathbf{A})^T\right\|_F^2+\frac{\mu}{2}\|\mathbf{J}-\mathbf{CW}\|_F^2\tag{25}$$

where $\mathbf{J}=\mathbf{E}-\mathbf{e}_n\mathbf{t}^T+\mathbf{Y}+\mathbf{L}*\mathbf{M}+(\mathbf{P}/\mu)$. Taking the partial derivative with respect to $\mathbf{C}$ and setting it to zero result in

$$\mu(\mathbf{CW}-\mathbf{J})\mathbf{W}^T-2\left(\mathcal{X}_{(3)}-\mathbf{C}(\mathbf{B}\odot\mathbf{A})^T\right)(\mathbf{B}\odot\mathbf{A})=0.\tag{26}$$

Then, $\mathbf{C}^\star$ can be updated by

$$\begin{aligned}\mathbf{C}^\star=&\left(\mu\mathbf{JW}^T+2\mathcal{X}_{(3)}(\mathbf{B}\odot\mathbf{A})\right)\\&\times\left(\mu\mathbf{WW}^T+2(\mathbf{B}\odot\mathbf{A})^T(\mathbf{B}\odot\mathbf{A})\right)^+.\end{aligned}\tag{27}$$

3) *Updating* $\mathbf{W}$*:* By fixing all other variables to solve $\mathbf{W}$, the objective function (17) is transformed into a classic orthogonal Procrustes problem, which can be expressed as

$$\mathbf{W}^\star=\underset{\mathbf{W}^T\mathbf{W}=\mathbf{I}}{\operatorname{argmin}}\|\mathbf{J}-\mathbf{CW}\|_F^2.\tag{28}$$

The problem (28) can be addressed in closed form through singular value decomposition (SVD)

$$\mathbf{W}^\star=\mathbf{VU}^T\tag{29}$$

where

$$\mathbf{U}\Sigma\mathbf{V}^T=\mathbf{J}^T\mathbf{C}.\tag{30}$$

4) *Updating* $\mathbf{t}$*:* For variable $\mathbf{t}$, (17) can be transformed into the following equivalent problem when all other variables are fixed:

$$\mathbf{t}^\star=\underset{\mathbf{t}}{\operatorname{argmin}}\left\|\mathbf{e}_n\mathbf{t}^T-\mathbf{K}\right\|_F^2\tag{31}$$

where $\mathbf{K}=\mathbf{E}-\mathbf{CW}+\mathbf{Y}+\mathbf{L}\star\mathbf{M}+(\mathbf{P}/\mu)$. Then the solution to $\mathbf{t}$ is updated by:

$$\mathbf{t}^\star=\frac{\mathbf{K}^T\mathbf{e}_n}{n}.\tag{32}$$

5) *Updating* $\mathbf{E}$*:* When all the variables are fixed except $\mathbf{E}$, (17) can be transformed into the following problem:

$$\mathbf{E}^\star=\underset{\mathbf{E}}{\operatorname{argmin}}\ \lambda\|\mathbf{E}\|_{2,1}+\frac{\mu}{2}\|\mathbf{E}-\mathbf{H}\|_F^2\tag{33}$$

where $\mathbf{H}=\mathbf{CW}+\mathbf{e}_n\mathbf{t}^T-\mathbf{Y}-\mathbf{L}*\mathbf{M}-(\mathbf{P}/\mu)$. Then, $\mathbf{E}^\star$ can be obtained by the well-known vector-soft threshold [56], [57]

$$\mathbf{E}^\star=\left(\frac{\max\left\{\|\mathbf{H}(k,:)\|_2-\frac{\lambda}{\mu},0\right\}}{\max\left\{\|\mathbf{H}(k,:)\|_2-\frac{\lambda}{\mu},0\right\}+\frac{\lambda}{\mu}}\right)\mathbf{H}\tag{34}$$

where $\mathbf{H}(k,:)$ denotes the kth row of $\mathbf{H}$.

6) *Updating* $\mathbf{M}$*:* By fixing all variables other than $\mathbf{M}$, the problem (17) can be transformed into the following optimization problem:

$$\mathbf{M}^\star=\underset{\mathbf{M}>0}{\operatorname{argmin}}\|\mathbf{Q}-\mathbf{L}*\mathbf{M}\|_F^2\tag{35}$$

where $\mathbf{Q}=\mathbf{CW}+\mathbf{e}_n\mathbf{t}^T-\mathbf{Y}-(\mathbf{P}/\mu)-\mathbf{E}$. It is well-known that the squared Frobenius norm of a matrix can be decomposed into the sum of the squares of its individual

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elements. Therefore, problem (35) can be decomposed into many subproblems as follows:

$$\mathbf{M}_{ij}^{\star} = \underset{\mathbf{M}_{ij}>0}{\operatorname{argmin}} \left(\mathbf{Q}_{ij} - \mathbf{L}_{ij} * \mathbf{M}_{ij}\right)^2 \tag{36}$$

where $\mathbf{Q}_{ij}$ $\mathbf{M}_{ij}$ and $\mathbf{L}_{ij}$ are the ijth elements of $\mathbf{Q}$, $\mathbf{M}$ and $\mathbf{L}$, respectively. Since $\mathbf{L}_{ij}^2 = 1$

$$\left(\mathbf{Q}_{ij} - \mathbf{L}_{ij} * \mathbf{M}_{ij}\right)^2 = \left(\mathbf{L}_{ij} * \mathbf{Q}_{ij} - \mathbf{M}_{ij}\right)^2. \tag{37}$$

Therefore, $\mathbf{M}^{\star}$ can be computed as follows:

$$\mathbf{M}^{\star} = \max(\mathbf{L} * \mathbf{Q}, \mathbf{0}). \tag{38}$$

7) *Updating* $\mathbf{P}$ *and* μ*:* For each iteration, $\mathbf{P}$ and μ are updated as follows:

$$\mathbf{P} \longleftarrow \mathbf{P} + \left(\mathbf{E} - \mathbf{C}\mathbf{W} - \mathbf{e}_n\mathbf{t}^T + \mathbf{Y} + \mathbf{L} * \mathbf{M}\right) \tag{39}$$

$$\mu \longleftarrow \rho\mu \tag{40}$$

where ρ is the incremental step size parameter.

The entire ADMM process is summarized in Algorithm 1.

Algorithm 1 ADMM

Input: The training data $\mathcal{X}$, the label matrix $\mathbf{Y}$, the luxury matrix $\mathbf{L}$, the regularization parameter λ and rank of tensor CP decomposition r.

Initiation: Initialize $\mathbf{W}$, $\mathbf{E}$, $\mathbf{M}$, $\mathbf{P}$, $\mathbf{t}$, as zero matrix or vector, respectively. Initialize $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, as identity matrix, respectively; $\mu = 1e^{-2}$, $\rho = 1.01$, $\epsilon = 1e^{-6}$.

repeat

Update $\mathbf{A}$, $\mathbf{B}$ by computing (24);
Update $\mathbf{C}$ by computing (27);
Update $\mathbf{W}$ by computing (29);
Update $\mathbf{t}$ by computing (32);
Update $\mathbf{E}$ by computing (34);
Update $\mathbf{M}$ by computing (38);
Update $\mathbf{P}$ and μ according to (39) and (40);

until $\|\mathbf{E} - \mathbf{C}\mathbf{W} - \mathbf{e}_n\mathbf{t}^T + \mathbf{Y} + \mathbf{L} * \mathbf{M}\|_F^2 < \epsilon$;

Output: The spatial dictionary $\mathbf{A}$, spectral dictionary $\mathbf{B}$ and transformation matrix $\mathbf{W}$.

C. Classification

When Algorithm 1 is completed, the transformation matrix $\mathbf{W}$, spatial dictionary $\mathbf{A}$ and spectral dictionary $\mathbf{B}$ are obtained. The label predictions for testing samples can be obtained in a simple two-step procedure. Suppose $\mathcal{X}_{\text{test}}$ is the testing sample set, then the double dictionary representation of testing samples can be first obtained by solving the following minimizing problem:

$$\mathbf{C}_{\text{test}}^{\star} = \underset{\mathbf{C}_{\text{test}}}{\operatorname{argmin}} \left\| \mathcal{X}_{\text{test}} - \mathbf{C}_{\text{test}}(\mathbf{B} \odot \mathbf{A})^T \right\|_F^2. \tag{41}$$

Taking the partial derivative with respect to $\mathbf{C}_{\text{test}}$ and setting it to zero, the optimal solution $\mathbf{C}_{\text{test}}^{\star}$ can be calculated by

$$\mathbf{C}_{\text{test}}^{\star} = \mathcal{X}_{\text{test}}(\mathbf{B} \odot \mathbf{A})\left(\left(\mathbf{B}^T\mathbf{B}\right) * \left(\mathbf{A}^T\mathbf{A}\right)\right)^{+}. \tag{42}$$

Then, the label matrix $\mathbf{Y}_{\text{test}}$ of the testing set can be simply predicted by

$$\mathbf{Y}_{\text{test}} = \mathbf{C}_{\text{test}}^{\star}\mathbf{W}. \tag{43}$$

TABLE I

AVERAGE COMPUTATIONAL TIME (IN SECONDS) OF DIFFERENT METHODS FOR FOUR DATASETS

Method	Salinas	Heihe	WHU-Hi-LongKou	WHU-Hi-Hanchuan
PCA	8.45	11.07	8.03	19.21
LPP	12.17	14.30	9.33	27.84
3D-CNN	317.34	241.08	180.77	402.63
GCN	182.06	202.53	189.12	298.22
OAR	16.73	15.30	12.54	20.03
BGCNN	157.90	175.74	124.87	193.25
DBMST	57.28	63.97	49.23	79.43
TDRLR	13.62	15.12	11.97	24.39

TABLE II

NUMBER OF TRAINING AND TEST SAMPLES OF THE SALINAS DATASET

	Class	Number of Samples	
No.	**Name**	**Training**	**Test**
1	Brocoli Green Weeds 1	20	1989
2	Brocoli Green Weeds 2	20	3689
3	Fallow	20	1957
4	Fallow Rough Plow	20	1381
5	Fallow Smooth	20	2652
6	Stubble	20	3920
7	Celery	20	3544
8	Grapes Untrained	20	11159
9	Soil Vinyard Develop	20	6141
10	Corn Senesced Green Weeds	20	3246
11	Lettuce Romaine 4wk	20	1058
12	Lettuce Romaine 5wk	20	1908
13	Lettuce Romaine 6wk	20	906
14	Lettuce Romaine 7wk	20	1060
15	Vinyard Untrained	20	7196
16	Vinyard Vertical Trellis	20	1789
	Total	320	53595

D. Computational Complexity

Let r be the rank of tensor $\mathcal{X} \in \mathbb{R}^{m\times l\times n}$, and c denotes the total number of classes. The complexity of computing $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ in each iteration are $O(r^3 + lmrn + jnr))$, $O(r^3 + lmrn + lnr))$, and $O(r^3 + lmrn + nr^2))$, respectively. For updating $\mathbf{W}$, the computational complexity can be divided into two parts: calculating $\mathbf{J}^{\mathbf{T}}\mathbf{C}$ and its SVD decomposition. The complexity for computing $\mathbf{J}^{\mathbf{T}}\mathbf{C}$ is $O(ncr))$ and the cost of its SVD are $O(r^3)$. The complexity of computing $\mathbf{t}$ is $O(nc)$. For updating $\mathbf{E}$, the computational complexity is only $O(nc)$ because the matrix $\mathbf{E}$ can be updated row by row. Updating $\mathbf{M}$ can be divided into $n \times c$ subproblems, and the computational complexity is also $O(nc)$. Since n is the number of training samples, which is generally much larger than the rank r and other variables. When the algorithm iterates for T rounds, the overall computational complexity is $O(T(n(lmr + lr + mr + r^2 + c)))$. Since a small training set is used during HSI training, the overall computational cost is not high.

To further illustrate, all methods were run ten times, and the average computational time is presented in Table I. It can be observed that the proposed TDRLR method exhibits significantly lower computational time compared to deep learning approaches, while being comparable to traditional methods such as PCA, LPP, and OAR. In conclusion, the running time of TDRLR is efficient and acceptable.

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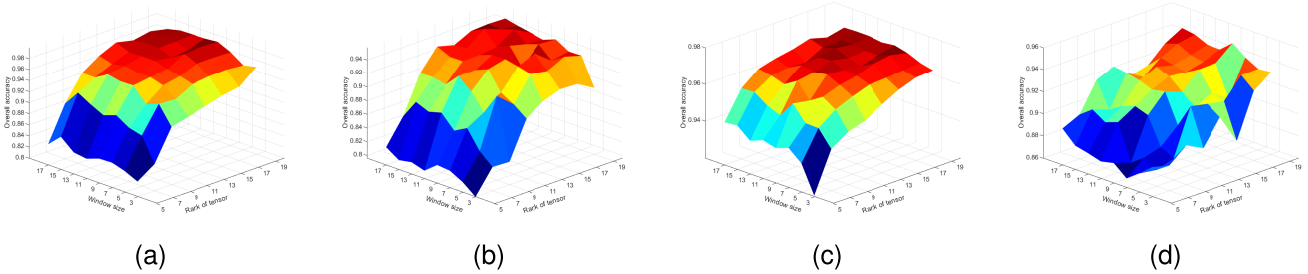


Fig. 3. Overall accuracy obtained by the proposed TDRLR with respect to the parameters w and r. (a) Salinas Valley. (b) Hei-He. (c) WHU-Hi-LongKou. (d) WHU-Hi-HanChuan.

TABLE III
NUMBER OF TRAINING AND TEST SAMPLES OF THE HEI-HE DATASET

	Class	Number of Samples	
No.	**Name**	**Training**	**Test**
1	Corn	20	41051
2	Fragrant-flowered Garlic	20	28566
3	Cauliflower	20	20335
4	Bell Pepper	20	7588
5	Potato	20	3742
6	Endive Sprout	20	1655
7	Watermelon	20	965
8	Artificial Surface	20	855
	Total	160	104757

TABLE IV
NUMBER OF TRAINING AND TEST SAMPLES OF WHU-HI-LONGKOU DATASET

	Class	Number of Samples	
No.	**Name**	**Training**	**Test**
1	Corn	25	34511
2	Cotton 2	25	8374
3	Sesame	25	3031
4	Broad-leaf soybean	25	63212
5	Narrow-leaf soybean	25	4151
6	Rice	25	11854
7	Water	25	67056
8	Roads and houses	25	7124
9	Mixed weed	25	5229
	Total	225	53595

TABLE V
NUMBER OF TRAINING AND TEST SAMPLES OF THE WHU-HI-HANCHUAN DATASET

	Class	Number of Samples	
No.	**Name**	**Training**	**Test**
1	Strawberry	50	44735
2	Cowpea	50	22753
3	Soybean	50	10287
4	Sorghum	50	5353
5	Water spinach	50	1200
6	Watermelon	50	4533
7	Greens	50	5903
8	Trees	50	17978
9	Grass	50	9469
10	Red roof	50	10516
11	Gray roof	50	16911
12	Plastic	50	3679
13	Bare soil	50	9116
14	Road	50	18560
15	Bright object	50	1136
16	Water	50	75401
	Total	800	257541

TABLE VI
HYPERPARAMETERS USED IN TDRLR FOR THE FOUR DATASETS

	w	r	λ	ρ	μ	ϵ
Salinas	15	17	$1e^{-1}$	1.01	$1e^{-2}$	$1e^{-6}$
Hei-He	13	9	$1e^{-1}$	1.01	$1e^{-2}$	$1e^{-6}$
WHU-Hi-LongKou	13	13	$1e^{-1}$	1.01	$1e^{-2}$	$1e^{-6}$
WHU-Hi-HanChuan	15	17	$1e^{-1}$	1.01	$1e^{-2}$	$1e^{-6}$

IV. EXPERIMENTS AND DISCUSSION

In this section, we compare the performance of the proposed method with seven state-of-the-art classification methods using four public HSI datasets. The experiments are conducted in MATLAB on a laptop equipped with an Intel i7 2.9-GHz CPU and 24 GB of memory.

A. Data Description

1) *Salinas:* This HSI was captured using the 224-band AVIRIS sensor, which offers high spatial resolution with 3.7-m pixels [58]. The dataset covers an area of 512 lines by 217 samples. 20 water absorption bands were excluded from the data, i.e., bands [108–112], [154–167], and 224. This imagery includes at-sensor radiance data and features a variety of surfaces, such as vegetables, bare soils, and vineyard fields. The ground truth for the Salinas dataset encompasses 16 distinct classes. The training set comprises 20 randomly selected labeled samples from each class. The quantities of training and testing samples are detailed in Table II.
2) *Hei-He:* The second HSI was collected using the CASI/SASI sensors in Zhangye Basin, located in the middle reaches of Heihe Watershed in Gansu Province, China [59]. The dataset has a size of 684 × 453 pixels and includes eight different types of land cover. After removing 14 bands that were heavily impacted by noise, the experiments were conducted using the remaining 135 bands. Table III provides more detailed information regarding the number of training and testing samples.
3) *WHU-Hi-LongKou:* The third dataset was collected on July 17, 2018, in Longkou Town, Hubei Province, China, using a Headwall Nano-Hyperspec imaging sensor

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TABLE VII
CLASSIFICATION RESULTS (%) OF DIFFERENT METHODS ON THE SALINAS DATASET

CL.#	PCA [10]	LPP [12]	3D-CNN [29]	GCN [39]	OAR [49]	BGCNN [44]	DBMST [48]	TDRLR
1	75.15±3.18	98.50±0.21	95.65±0.78	97.45±0.21	98.51±0.74	96.67±3.67	98.28±1.79	**98.97±0.37**
2	95.24±8.70	98.57±0.35	97.23±0.25	99.04±0.45	98.97±1.63	93.97±2.57	98.04±1.29	**99.12±0.49**
3	84.54±3.33	95.54±2.97	90.87±5.39	82.14±0.96	**99.80±0.31**	98.62±4.29	98.79±0.66	92.79±0.05
4	98.85±0.48	90.32±1.24	97.45±2.39	83.92±5.24	98.52±0.54	98.87±0.65	99.55±0.10	**100.00±0.00**
5	86.24±1.52	96.02±1.51	87.12±2.98	90.45±5.01	88.23±1.07	90.11±3.68	97.59±0.29	**97.70±1.28**
6	90.33±0.31	91.27±0.02	82.23±3.96	99.12±0.48	89.52±2.44	95.03±3.49	99.20±0.75	**99.61±0.15**
7	95.54±1.98	82.54±0.14	90.87±0.23	98.34±0.65	95.94±3.15	95.21±2.48	99.57±0.31	**100.00±0.00**
8	79.25±2.48	85.25±2.84	80.25±4.29	86.14±8.87	86.94±0.43	94.60±1.76	86.27±2.79	**98.05±0.38**
9	97.56±0.41	86.98±1.09	88.35±0.14	91.78±0.40	99.11±2.73	92.23±6.81	99.03±0.06	**100.00±0.00**
10	81.83±5.90	93.14±2.78	92.14±5.12	94.99±0.61	97.02±1.90	**97.14±5.01**	96.35±1.39	93.31±0.31
11	89.18±2.15	**99.45±0.17**	94.65±1.23	96.03±1.28	99.33±0.69	80.95±2.53	94.67±3.94	91.16±0.10
12	94.12±6.02	94.65±0.94	89.96±1.78	96.45±2.21	95.34±1.45	85.81±2.85	96.05±2.57	**96.87±0.01**
13	90.31±1.05	93.12±0.94	**96.74±4.15**	94.17±1.84	89.43±6.32	80.54±3.58	94.88±1.59	95.91±0.02
14	81.22±1.52	85.07±2.31	70.56±0.41	68.02±1.98	95.47±2.21	87.88±2.15	**97.64±2.53**	90.06±0.09
15	80.01±3.64	79.43±1.62	87.14±0.78	85.55±1.98	92.80±0.69	83.54±3.85	85.23±5.64	**93.69±0.78**
16	85.21±1.16	92.98±0.81	91.47±2.51	91.74±0.44	89.05±2.74	92.28±1.97	94.13±2.57	**96.00±3.12**
OA	88.15±2.04	90.23±2.17	90.73±1.98	90.32±1.56	95.04±1.32	93.92±3.26	94.35±1.54	**97.07±0.19**
AA	86.90±2.35	91.15±1.28	88.76±2.45	91.19±1.71	94.27±1.25	93.76±3.02	94.47±1.55	**95.97±0.07**
κ	81.87±2.06	88.69±2.87	87.06±0.87	85.55±3.25	90.16±1.51	92.15±3.17	94.04±0.97	**96.44±0.15**

TABLE VIII
CLASSIFICATION RESULTS (%) OF DIFFERENT METHODS ON THE HEIHE DATASET

CL.#	PCA [10]	LPP [12]	3D-CNN [29]	GCN [39]	OAR [49]	BGCNN [44]	DBMST [48]	TDRLR
1	90.30±1.56	83.94±1.75	88.55±1.91	83.04±1.06	**95.91±2.14**	86.21±1.54	86.67±1.93	93.09±1.58
2	92.51±0.41	92.51±0.64	90.63±0.31	86.87±0.34	96.26±1.51	93.82±1.06	98.18±0.72	**99.13±1.18**
3	78.71±8.52	86.19±2.87	57.28±2.64	90.37±0.02	85.22±1.54	91.60±5.21	94.13±2.53	**94.53±2.34**
4	93.21±0.24	90.77±3.42	93.90±0.33	89.75±7.07	84.40±0.80	91.20±1.25	89.57±4.72	**94.86±0.57**
5	97.80±0.66	**99.71±0.14**	98.85±1.72	97.64±5.23	91.82±0.54	95.69±0.59	93.14±2.24	98.12±2.60
6	79.54±3.64	92.38±1.43	74.52±5.88	80.37±0.98	95.91±0.32	84.09±0.62	92.89±0.95	**97.18±1.22**
7	86.91±9.60	85.54±2.06	71.59±4.84	**93.80±3.21**	90.73±0.95	89.69±1.54	86.71±1.84	87.88±0.81
8	88.11±3.01	84.04±2.93	76.81±4.07	75.34±4.54	96.79±0.67	90.15±1.14	87.28±3.98	**98.08±1.21**
OA	88.01±1.54	85.08±2.70	84.31±1.67	87.15±2.07	92.99±0.23	90.38±0.67	91.81±2.46	**94.71±1.54**
AA	87.74±3.27	89.70±1.74	86.36±2.57	91.78±2.50	92.29±1.08	92.47±1.60	91.07±2.52	**94.05±1.43**
κ	84.18±2.03	85.31±3.94	79.53±2.21	87.71±2.28	90.18±1.61	88.52±2.12	89.01±1.97	**92.25±0.98**

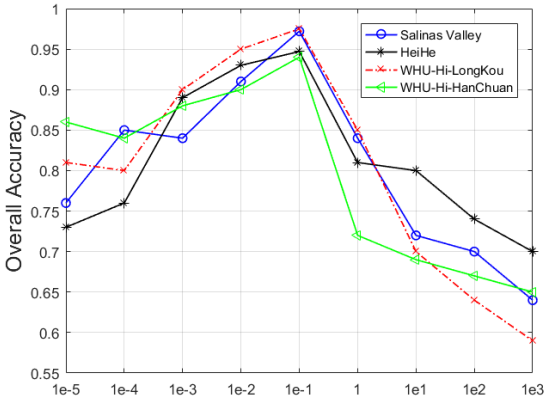


Fig. 4. Overall accuracy obtained by the proposed TDRLR with respect to the parameter λ. (a) Salinas Valley. (b) Hei-He. (c) WHU-Hi-LongKou. (d) WHU-Hi-HanChuan.

mounted on a DJI Matrice 600 Pro UAV platform [60]. Data was acquired under clear skies, with a temperature of approximately 36 °C and a relative air humidity of about 65%. The study area is an agricultural landscape containing six crop species: corn, cotton, sesame, broad-leaf soybean, narrow-leaf soybean, and rice. The UAV flew at an altitude of 500 m, capturing imagery with a size of 550 by 400 pixels across 270 bands, ranging from 400 to 1000 nm. The spatial resolution of UAV-borne HSI is approximately 0.463 m. The numbers of training and testing samples are summarized in Table IV.

4) *WHU-Hi-HanChuan:* The WHU-Hi-HanChuan dataset was collected on June 17, 2016, in Hanchuan, Hubei Province, China. Data acquisition was performed using a Headwall Nano-Hyperspec imaging sensor with a 17-mm focal length, mounted on a Leica Aibot X6 UAV V1 platform [60]. The weather was clear and cloudless, with a relative air humidity of about 70%. The study area encompasses a rural-urban fringe zone featuring buildings, water bodies, and cultivated land. It includes seven crop species: strawberry, cowpea, soybean, sorghum, water spinach, watermelon, and greens. The numbers of training and testing samples are detailed in Table V.

B. Experimental Setup

To verify the effectiveness of the proposed TDRLR method, we compared it against seven methods: PCA [10], LPP [12], 3D-CNN [29], GCN [39], OAR [49], BGCNN [44], and DBMST [48]. Among these, LPP, PCA, and OAR are popular subspace learning methods, while the remaining four—

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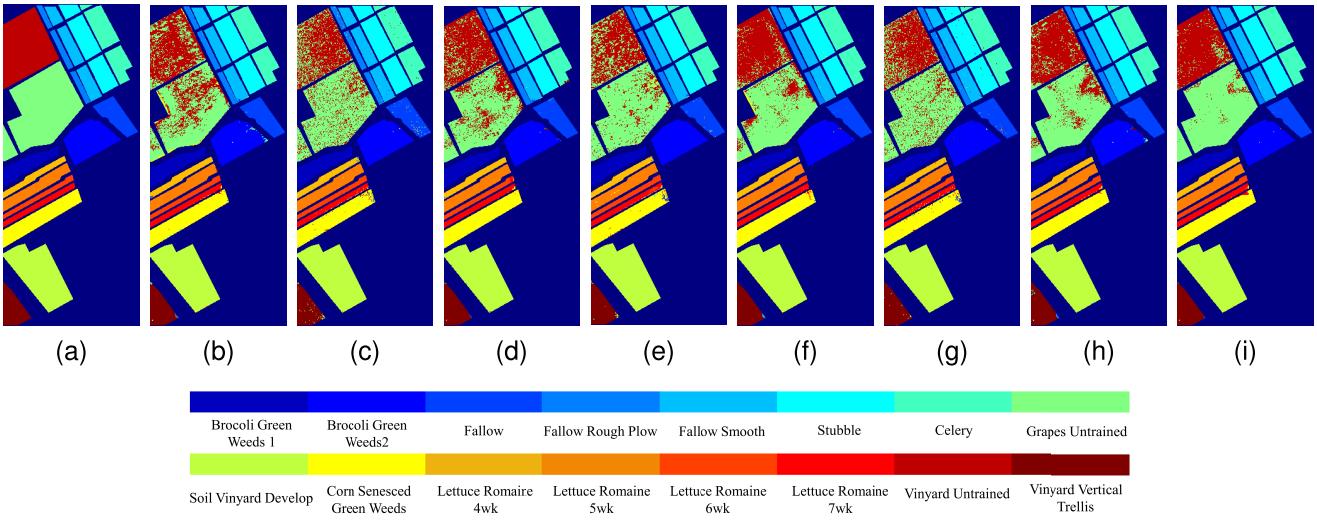


Fig. 5. Classification maps of different methods for the Indian Pines dataset. (a) Ground truth. (b) PCA. (c) LPP. (d) 3D-CNN. (e) GCN. (f) OAR. (g) BGCNN. (h) DBMST. (i) TDRLR.

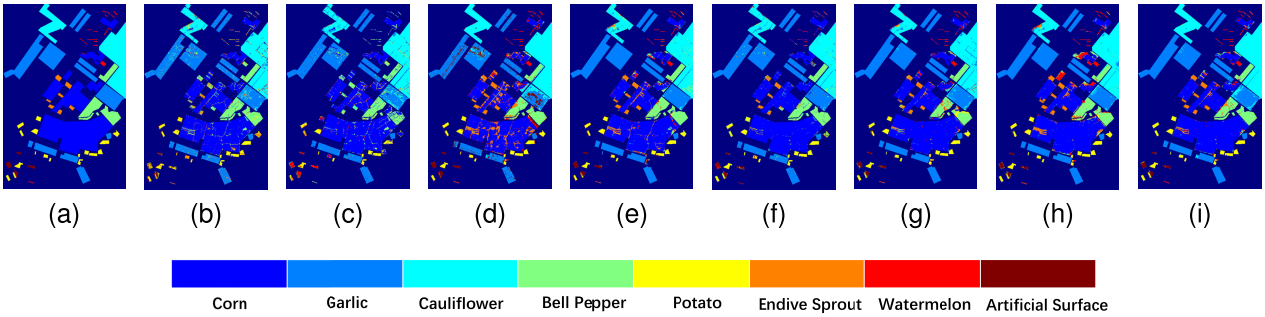


Fig. 6. Classification maps of different methods for the Hei-He dataset. (a) Ground truth. (b) PCA. (c) LPP. (d) 3D-CNN. (e) GCN. (f) OAR. (g) BGCNN. (h) DBMST. (i) TDRLR.

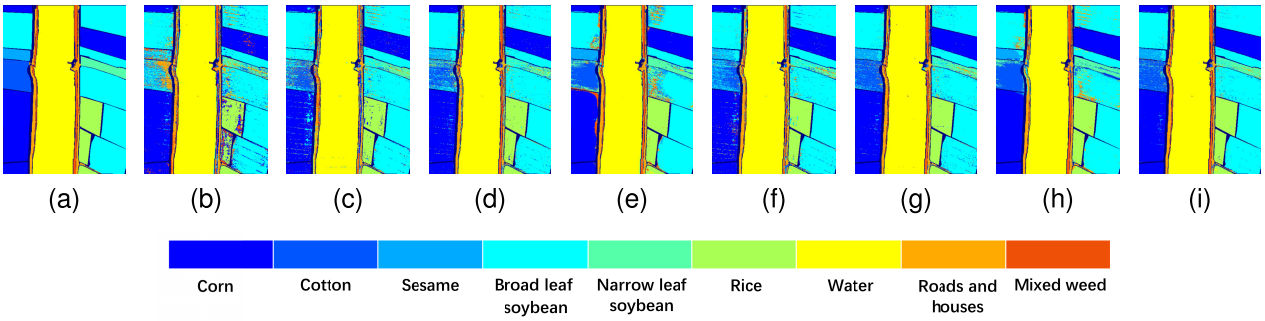


Fig. 7. Classification maps of different methods for the WHU-Hi-LongKou dataset. (a) Ground truth. (b) PCA. (c) LPP. (d) 3D-CNN. (e) GCN. (f) OAR. (g) BGCNN. (h) DBMST. (i) TDRLR.

3D-CNN, GCN, BGCNN, and DBMST—are deep learning methods. Each experiment was run independently ten times, and the average results were reported. The parameters for the comparative experiments were set as follows.

1) For PCA, OAR, and LPP, the reduced dimension is set to 30. To fully utilize spatial information across all bands, the SVM-CK [61] classifier is utilized for classification.
2) For BGCNN, the group of feature nodes is set to 10 and the group of enhancement nodes is set to 300. In GCN, the same network configuration is utilized as in [39].

For the proposed TDRLR method, three parameters need to be determined: the window size w, the rank of the tensor decomposition r, and the regularization parameter λ. Since the proposed method uses spatial–spectral tensor samples, an appropriate w can significantly impact the quality of the classification results, while r influences the representational ability of subspace learning. In our experiment, w is chosen from the range of $\{3, 5, 7, 9, 11, 13, 15, 17\}$, and r is selected from $\{3, 5, 7, 9, 11, 13, 15, 17\}$. Fig. 3 shows the overall accuracy obtained by the proposed TDRLR with respect to w and r. It can be seen that smaller values of w and r fail to achieve high accuracy. However, increasing the values of w and r leads to higher complexity. After careful evaluation, the values for $\{w, r\}$ in TDRLR are set to $\{15, 17\}$,

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TABLE IX
CLASSIFICATION RESULTS (%) OF DIFFERENT METHODS ON THE WHU-HI-LONGKOU DATASET

CL.#	PCA [10]	LPP [12]	3D-CNN [29]	GCN [39]	OAR [49]	BGCNN [44]	DBMST [48]	TDRLR
1	90.03±1.65	85.94±1.75	88.55±1.91	83.04±1.06	95.87±2.12	96.21±1.54	**99.36±0.76**	93.09±1.58
2	91.51±0.64	90.13±0.64	90.63±0.31	86.87±0.34	94.62±1.25	93.82±1.06	89.06±2.90	**95.13±1.18**
3	78.71±8.52	86.91±2.87	97.28±2.64	90.37±0.02	87.33±1.71	94.60±5.21	86.97±3.09	**97.53±2.34**
4	93.12±0.24	90.27±2.24	94.90±0.33	89.75±7.07	92.25±0.80	96.20±1.25	89.72±1.29	**98.86±0.57**
5	96.51±0.66	98.71±0.14	**98.85±1.72**	98.64±5.23	89.82±0.54	95.69±0.59	96.44±1.71	97.12±2.60
6	77.45±3.61	92.35±1.34	74.52±5.88	90.37±0.98	95.15±0.44	84.09±0.62	93.17±2.13	**97.18±1.22**
7	72.84±9.60	78.54±2.06	71.59±4.84	93.80±3.21	91.73±0.59	96.69±1.54	95.99±1.03	**97.88±0.81**
8	88.11±3.01	84.04±2.93	76.81±4.07	95.34±4.54	96.79±0.67	90.15±1.14	90.82±2.95	**98.08±1.21**
9	94.87±1.51	99.12±0.45	97.12±1.44	**99.82±0.03**	94.77±1.18	82.75±1.43	88.16±1.43	90.96±2.26
OA	86.71±3.27	89.70±1.74	87.36±2.57	92.33±2.50	94.90±1.08	92.63±1.60	94.74±1.64	**97.31±1.53**
AA	87.64±1.54	88.08±5.70	88.31±1.67	87.15±2.07	92.85±0.21	93.38±0.67	92.64±1.74	**96.67±1.54**
κ	83.97±2.03	84.63±3.94	88.53±2.21	87.71±2.28	90.88±1.61	88.52±2.12	93.17±1.56	**93.25±0.98**

TABLE X
CLASSIFICATION RESULTS (%) OF DIFFERENT METHODS ON THE WHU-HI-HANCHUAN DATASET

CL.#	PCA [10]	LPP [12]	3D-CNN [29]	GCN [39]	OAR [49]	BGCNN [44]	DBMST [48]	TDRLR
1	85.21±0.69	71.95±1.73	91.81±1.07	89.08±1.51	93.57±0.74	96.07±3.79	94.95±1.03	**97.13±1.23**
2	70.31±7.20	65.59±2.73	79.83±3.65	62.29±4.25	81.68±1.25	85.58±4.37	86.80±1.67	**95.28±1.15**
3	73.43±3.12	91.65±1.51	92.08±4.23	94.90±1.96	79.46±0.63	80.82±4.13	90.68±3.27	**96.38±0.70**
4	77.23±4.92	87.80±4.96	92.51±1.63	95.51±0.64	85.21±0.33	83.18±1.18	98.96±0.81	**99.19±0.12**
5	88.51±4.71	91.59±2.40	**99.81±2.43**	76.21±7.21	82.41±5.98	81.51±3.91	98.77±0.18	91.50±2.33
6	84.33±1.62	75.32±1.25	65.71±3.11	72.69±3.43	76.03±2.17	84.34±1.83	71.87±5.60	**89.17±3.10**
7	87.05±2.50	70.97±6.86	97.93±1.46	**97.96±0.65**	93.06±3.48	87.50±2.61	95.96±1.71	89.92±2.53
8	81.25±5.27	84.25±2.54	77.25±2.18	83.10±4.04	84.27±1.15	**96.80±1.39**	86.37±2.46	93.07±1.59
9	80.25±1.86	77.35±1.19	84.47±3.14	89.76±3.27	85.06±2.41	90.23±0.37	87.54±4.73	**92.59±1.23**
10	96.14±0.26	92.41±3.31	88.75±5.02	82.32±1.26	88.38±1.10	95.66±1.99	92.19±3.25	**99.21±0.41**
11	80.16±2.64	90.01±4.59	92.03±0.24	91.73±0.91	91.05±0.52	96.75±2.53	83.37±4.64	**97.75±0.75**
12	92.57±1.69	94.26±0.74	97.06±0.57	90.93±3.06	78.46±1.55	**99.35±2.70**	98.06±1.59	90.61±4.94
13	80.38±2.70	80.96±1.73	**90.35±0.81**	86.55±2.90	82.79±2.03	80.54±1.29	82.46±8.17	82.32±4.78
14	78.16±5.13	95.11±2.02	83.77±0.71	95.74±4.37	80.77±2.58	93.74±0.16	88.27±5.27	**97.90±2.80**
15	92.06±6.11	96.96±2.68	93.55±2.68	96.31±1.61	72.33±5.79	96.53±0.85	**98.56±0.94**	82.35±4.22
16	93.21±4.12	99.26±0.01	96.64±0.79	93.97±0.74	99.22±0.28	97.28±0.43	95.79±2.08	**99.74±0.05**
OA	80.08±2.79	83.43±2.18	90.13±2.45	85.19±1.71	87.90±0.85	90.41±0.85	91.56±2.31	**94.02±0.19**
AA	85.25±3.15	86.83±2.58	88.60±2.11	86.10±2.61	84.27±2.03	90.08±2.18	90.66±2.26	**91.95±0.92**
κ	79.26±2.36	80.40±2.87	88.19±0.87	84.84±3.25	82.41±2.01	89.15±2.56	90.01±1.39	**92.64±0.15**

$\{13, 9\}$, $\{13, 13\}$, $\{15, 17\}$ across the Salinas, Hei-He, WHU-Hi-LongKou, and WHU-Hi-HanChuan datasets, respectively. The parameter λ controls the balance between the double dictionary representation error and the relaxation regression error. The value of λ is selected experimentally from the range $\{1e^3, 1e^2, 1e^1, 1e^{-1}, 1e^{-2}, 1e^{-3}, 1e^{-4}, 1e^{-5}\}$. Fig. 4 illustrates the overall accuracy obtained by the proposed TDRLR with respect to parameter λ. It is evident that when λ is around $1e^{-1}$, the classification accuracy is higher. Based on the observation, a common parameter setting for all four HSI datasets is $\lambda = \{0.1\}$. All hyperparameters of the proposed TDRLR are listed in Table VI.

C. Classification Performance

Tables VII–X present the per-class accuracy, overall accuracy (OA), average accuracy (AA), and Kappa coefficient (κ) for various methods.

Table VII details the experimental results for the Salinas Valley dataset. The proposed TDRLR method demonstrates a substantial improvement in OA, outperforming PCA, LPP, 3D-CNN, GCN, OAR, BGCNN, and DBMST by 8.92%, 6.64%, 6.34%, 6.75%, 2.03%, 3.15%, and 2.72%, respectively. In addition, TDRLR shows noticeable improvements in both AA and κ. A particularly significant observation is the per-class accuracy from the TDRLR with a limited number of labeled training samples. For instance, for class 8, with only 20 training samples and 11 159 testing samples, and for class 15, with 20 training samples and 7196 testing samples, TDRLR offers the highest per-class accuracy of 98.05% and 93.69%, respectively, despite the substantial disparity between the number of training and test samples. Tables VIII–X present accuracy assessments for the HeiHe dataset, the WHU-Hi-LongKou dataset, and the WHU-Hi-Hanchuan dataset, respectively. The results consistently show that TDRLR outperforms the other methods, confirming its effectiveness across diverse HSI datasets.

Among the four datasets, the WHU-Hi-LongKou dataset exhibits the best classification performance. This is likely due to the higher clustering degree of each class in this dataset, allowing for effective classification with a relatively small number of samples. In contrast, the WHU-Hi-HanChuan dataset presents more challenges for prediction. This is because it contains a greater number of classes, and the spatial distances between these classes are closer, making it harder to differentiate between them. Notably, when the number of training samples per category is consistent across

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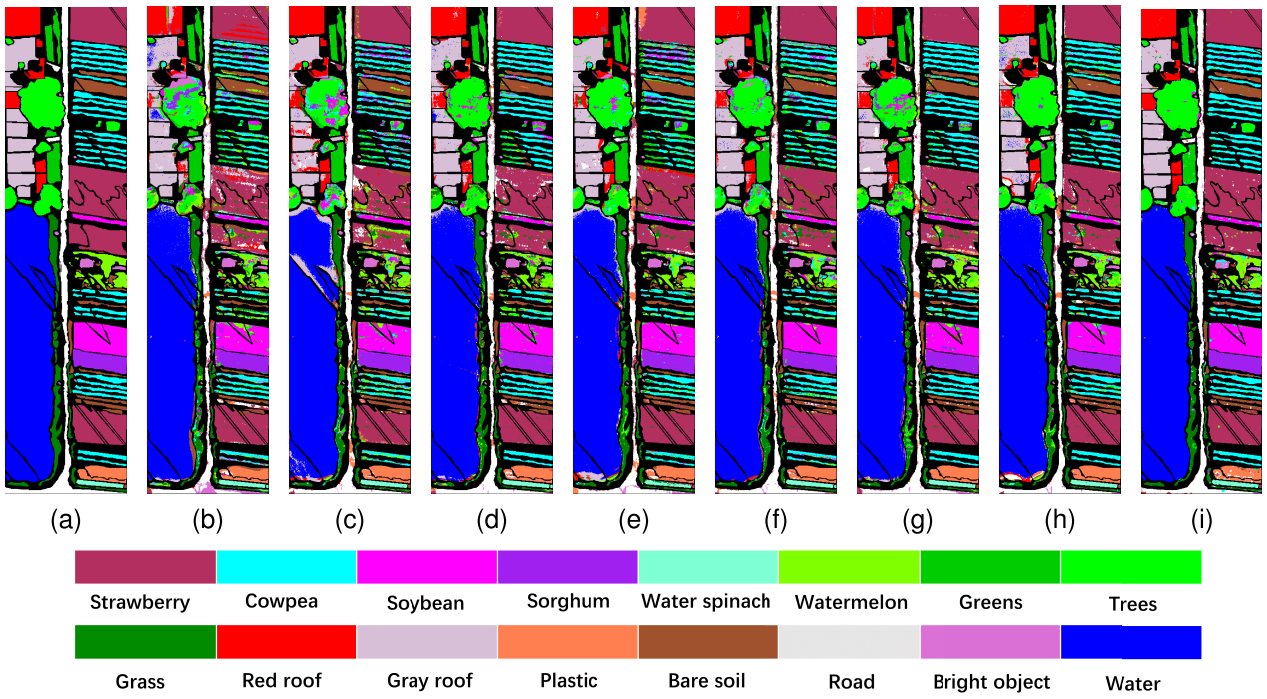


Fig. 8. Classification maps of different methods for the WHU-HI-HanChuan dataset. (a) Ground truth. (b) PCA. (c) LPP. (d) 3D-CNN. (e) GCN. (f) OAR. (g) BGCNN. (h) DBMST. (i) TDRLR.

TABLE XI
ABLATION STUDY OF THE SALINAS DATASET

Baseline	✓	✓	✓	✓
TD	✗	✓	✗	✓
LR	✗	✗	✓	✓
OA	65.68±0.45	93.63±2.37	79.98±1.46	**97.07±0.19**
AA	59.36±0.46	91.56±1.11	78.95±0.97	**95.97±0.07**
κ	61.67±0.62	89.42±1.81	70.14±0.57	**96.44±0.15**

TABLE XII
ABLATION STUDY OF THE HEIHE DATASET

Baseline	✓	✓	✓	✓
TD	✗	✓	✗	✓
LR	✗	✗	✓	✓
OA	61.42±2.53	89.32±3.67	76.07±0.51	**94.71±1.54**
AA	59.51±0.69	86.89±2.73	73.18±1.76	**94.05±1.43**
κ	54.62±0.76	82.36±2.81	68.14±1.57	**91.25±0.98**

TABLE XIII
ABLATION STUDY OF THE WHU-HI-LONGKOU DATASET

Baseline	✓	✓	✓	✓
TD	✗	✓	✗	✓
LR	✗	✗	✓	✓
OA	72.25±2.63	94.36±0.36	86.32±0.69	**97.31±1.53**
AA	74.56±1.50	91.26±0.71	82.18±0.32	**96.67±1.54**
κ	71.90±3.79	89.78±0.53	81.55±1.13	**91.25±0.98**

datasets, the proposed TDRLR method achieves an OA on the WHU-Hi-LongKou dataset that is 3.29% higher than on the WHU-Hi-HanChuan dataset. This highlights the effectiveness of the TDRLR, particularly in datasets where class separability is more distinct.

TABLE XIV
ABLATION STUDY OF THE WHU-HI-HANCHUAN DATASET

Baseline	✓	✓	✓	✓
TD	✗	✓	✗	✓
LR	✗	✗	✓	✓
OA	54.38±2.58	84.59±2.88	69.97±1.54	**94.02±0.19**
AA	52.14±1.10	81.12±1.56	65.85±1.99	**91.95±0.92**
κ	50.26±1.96	79.26±1.27	63.31±2.05	**92.64±0.15**

It is evident that 3D-CNN, GCN, BGCNN, and DBMST achieve acceptable classification results due to their ability to effectively extract spatial–spectral information from HSI. However, their performance is limited in scenarios with small training sample sizes, as these methods typically require larger sample sets to achieve high training accuracy. Among the subspace learning methods such as LPP, PCA, OAR, and TDRLR, OAR demonstrates good classification results, mainly because it incorporates the concept of relaxation regression. However, on the WHU-Hi-HanChuan dataset, which is more challenging to classify, OAR performs worse than the deep learning methods. This is primarily because OAR, unlike deep learning approaches, cannot extract spatial–spectral information from HSI. In comparison, TDRLR significantly improves classification accuracy over OAR, with an increase of 2.03% in OA for the Salinas Valley dataset, 1.71% for the HeiHe dataset, 2.41% for the WHU-Hi-LongKou dataset, and 6.12% for the WHU-Hi-HanChuan dataset. These results demonstrate that the proposed TDRLR method is superior to traditional subspace learning methods, particularly in its ability to extract spatial–spectral information from HSI under conditions of limited training data.

Figs. 5–8 display the classification maps generated by various methods. From these figures, it is evident that the

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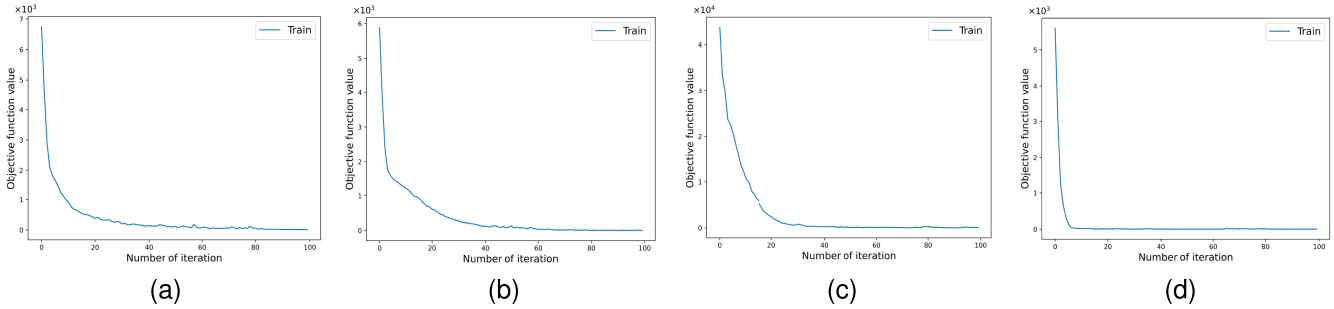


Fig. 9. Convergence of TDRLR on four datasets. (a) Salinas Valley. (b) Hei-He. (c) WHU-Hi-LongKou. (d) WHU-Hi-HanChuan.

TABLE XV
CLASSIFICATION RESULTS ON DIFFERENT DATASETS CORRUPTED BY GAUSSIAN NOISE

Method	Salinas Dataset			Hei-He Dataset			WHU-Hi-LongKou Dataset			WHU-Hi-HanChuan Dataset		
	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
TDRLR-F	78.84±2.29	75.94±2.06	73.40±1.38	75.39±1.77	74.14±1.90	72.20±2.60	80.95±2.89	79.17±3.76	74.67±2.45	72.43±4.54	70.93±2.39	69.76±3.06
TDRLR	**90.03±0.83**	**88.21±1.47**	**86.38±1.88**	**88.70±1.79**	**86.19±1.89**	**85.24±1.32**	**91.78±0.54**	**90.58±0.74**	**88.34±0.51**	**86.33±1.48**	**83.18±1.92**	**82.26±0.90**

proposed method delivers the smoothest classification results. This suggests that the TDRLR method not only accurately classifies HSI but also produces consistent and visually coherent classification maps, which are crucial for practical applications. Fig. 9 illustrates the behavior of our objective function across the four HSI datasets as the number of iterations increases. The objective function shows a steady decrease, with minimal changes observed after 50 iterations. This indicates that our iterative algorithm is highly efficient, ensuring that the model reaches a stable solution in a reasonable amount of time.

D. Ablation Study

In this section, we perform an ablation study to assess the contributions of the tensor decomposition term and LR to the overall performance of the model. A linear regression model serves as the baseline for comparison. ✓ indicates that the corresponding module is added to the baseline, and ✗ represents that the corresponding module is not included. We use the average OA, AA, and κ across different datasets to evaluate classification performance. To ensure a fair assessment, each method is executed ten times. The experimental results, presented in Tables XI–XIV, clearly demonstrate the significant impact of both tensor decomposition and LR on enhancing classification performance. For instance, in the Salinas dataset, as presented in Table XI, incorporating tensor decomposition into the baseline model results in substantial improvements, with approximately a 28% increase in OA, a 32% increase in AA, and a 28% increase in κ. Adding LR to the baseline also leads to considerable gains, including a 14% rise in OA, a 19% rise in AA, and an 18% rise in κ. Interestingly, the results suggest that tensor decomposition has a more pronounced effect on the model's performance compared to LR. Specifically, when comparing the baseline + tensor decomposition model with the proposed TDRLR model (baseline + tensor decomposition + LR), we observe additional improvements of 3.07% in OA, 4.18% in AA, and 4.78% in κ. Similar patterns are observed across the other three datasets, as detailed in Tables XII–XV. These findings underscore the importance of both tensor decomposition and LR in the TDRLR model, with each component playing a crucial role in boosting classification accuracy and robustness.

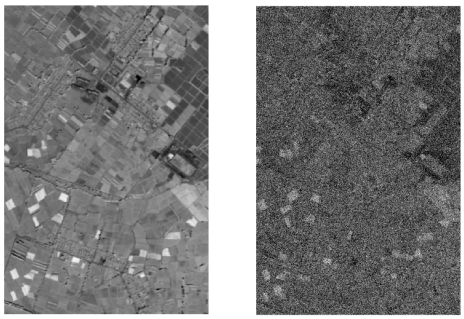


Fig. 10. Example image of Heihe dataset in 60th band. (a) Original image. (b) Image with Gaussian noise.

E. Robustness Analysis

As mentioned in Section III-A, the Frobenius norm is sensitive to outliers and performs poorly with noisy data, whereas the $L_{2,1}$ norm is more robust to such noise. To verify, the classification results of TDRLR and the Frobenius norm version of TDRLR (denoted as TDRLR-F) are compared. In the experiments, Gaussian noise is added to the four datasets, with each band being contaminated by zero-mean Gaussian noise with random intensities. The corresponding variances range between 0 and 0.5, and an example of a noisy band is shown in Fig. 10.

Table XV presents the classification results of the TDRLR-F and TDRLR on the noisy datasets. It can be observed that TDRLR consistently outperforms TDRLR-F. Specifically, on the Salinas dataset, TDRLR shows improvements of over 11.17% in OA, 12.27% in AA, and 12.98% in κ compared

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to TDRLR-F under Gaussian noise. Similar improvements are observed across the other three datasets as well. These results demonstrate that the use of the $L_{2,1}$ norm enhances the model's robustness to noisy data, improving classification performance.

V. Conclusion

This article introduced a novel method called TDRLR for the classification of HSIs. The method effectively integrates CP-based tensor regression into linear regression to obtain a robust double dictionary representation. In addition, it employs LR to increase interclass distance, thereby enhancing the discriminative power of the transformation matrix when mapping the subspace representation into the label space. Extensive experiments conducted on several HSI datasets demonstrated that the proposed TDRLR method outperformed traditional subspace learning methods as well as several state-of-the-art deep learning approaches, achieving superior classification accuracy and robustness. In the future work, we will make an effort to extend the method into an unsupervised or self-supervised manner for dealing with the classification of HSIs that lack labels.

References

[1] A. M. Wijata et al., "Taking artificial intelligence into space through objective selection of hyperspectral earth observation applications: To bring the 'brain' close to the 'eyes' of satellite missions," *IEEE Geosci. Remote Sens. Mag.*, vol. 11, no. 2, pp. 10–39, Jun. 2023.

[2] P. Duan, X. Kang, P. Ghamisi, and S. Li, "Hyperspectral remote sensing benchmark database for oil spill detection with an isolation forest-guided unsupervised detector," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5509711.

[3] J. Nalepa et al., "Estimating soil parameters from hyperspectral images: A benchmark dataset and the outcome of the HYPERVIEW challenge," *IEEE Geosci. Remote Sens. Mag.*, vol. 12, no. 3, pp. 35–63, Sep. 2024.

[4] L. Ren, L. Gao, M. Wang, X. Sun, and J. Chanussot, "HADGSM: A unified nonconvex framework for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5503415.

[5] C. Deepa, A. Shetty, and A. V. Narasimhadhan, "Performance evaluation of dimensionality reduction techniques on hyperspectral data for mineral exploration," *Earth Sci. Informat.*, vol. 16, no. 1, pp. 25–36, Feb. 2023.

[6] L. Ren, D. Hong, L. Gao, X. Sun, M. Huang, and J. Chanussot, "Orthogonal subspace unmixing to address spectral variability for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501713.

[7] L. Ren, D. Hong, L. Gao, X. Sun, M. Huang, and J. Chanussot, "Hyperspectral sparse unmixing via nonconvex shrinkage penalties," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5500415.

[8] B. F. R. Davies et al., "Multi- and hyperspectral classification of soft-bottom intertidal vegetation using a spectral library for coastal biodiversity remote sensing," *Remote Sens. Environ.*, vol. 290, May 2023, Art. no. 113554.

[9] L. Shuai, Z. Li, Z. Chen, D. Luo, and J. Mu, "A research review on deep learning combined with hyperspectral imaging in multiscale agricultural sensing," *Comput. Electron. Agricult.*, vol. 217, Feb. 2024, Art. no. 108577.

[10] Y. Wang, T. Li, L. Chen, Y. Yu, Y. Zhao, and J. Zhou, "Tensor-based robust principal component analysis with locality preserving graph and frontal slice sparsity for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5508319.

[11] S. D. Fabiyi, P. Murray, J. Zabalza, and J. Ren, "Folded LDA: Extending the linear discriminant analysis algorithm for feature extraction and data reduction in hyperspectral remote sensing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 12312–12331, 2021.

[12] H. Guo, H. Zou, and J. Tan, "Semi-supervised dimensionality reduction via sparse locality preserving projection," *Appl. Intell.*, vol. 50, no. 4, pp. 1222–1232, Apr. 2020.

[13] Z. Gong, W. Hu, X. Du, P. Zhong, and P. Hu, "Deep manifold embedding for hyperspectral image classification," *IEEE Trans. Cybern.*, vol. 52, no. 10, pp. 10430–10443, Oct. 2022.

[14] G. Taskin and G. Camps-Valls, "Graph embedding via high dimensional model representation for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5519811.

[15] N. H. Ly, Q. Du, and J. E. Fowler, "Sparse graph-based discriminant analysis for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 7, pp. 3872–3884, Jul. 2014.

[16] W. Li, J. Liu, and Q. Du, "Sparse and low-rank graph for discriminant analysis of hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 7, pp. 4094–4105, Jul. 2016.

[17] F. de Morsier, M. Borgeaud, V. Gass, J.-P. Thiran, and D. Tuia, "Kernel low-rank and sparse graph for unsupervised and semi-supervised classification of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 6, pp. 3410–3420, Jun. 2016.

[18] Y.-J. Deng, H.-C. Li, L. Pan, and W. J. Emery, "Tensor locality preserving projection for hyperspectral image classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2017, pp. 771–774.

[19] Y.-J. Deng, H.-C. Li, L. Pan, L.-Y. Shao, Q. Du, and W. J. Emery, "Modified tensor locality preserving projection for dimensionality reduction of hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 2, pp. 277–281, Feb. 2018.

[20] L. Pan, H.-C. Li, Y.-J. Deng, F. Zhang, X.-D. Chen, and Q. Du, "Hyperspectral dimensionality reduction by tensor sparse and low-rank graph-based discriminant analysis," *Remote Sens.*, vol. 9, no. 5, p. 452, May 2017.

[21] J. An, X. Zhang, H. Zhou, J. Feng, and L. Jiao, "Patch tensor-based sparse and low-rank graph for hyperspectral images dimensionality reduction," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 7, pp. 2513–2527, Jul. 2018.

[22] P. Wang, C. Zheng, and S. Xiong, "Hyperspectral image dimensionality reduction via graph embedding in core tensor space," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 3, pp. 509–513, Mar. 2021.

[23] S. Jin et al., "CSKNN: Cost-sensitive K-nearest neighbor using hyperspectral imaging for identification of wheat varieties," *Comput. Electr. Eng.*, vol. 111, Oct. 2023, Art. no. 108896.

[24] S. Fei, S. Xiao, J. Zhu, Y. Xiao, and Y. Ma, "Dual sampling linear regression ensemble to predict wheat yield across growing seasons with hyperspectral sensing," *Comput. Electron. Agricult.*, vol. 216, Jan. 2024, Art. no. 108514.

[25] G. Liu, L. Wang, D. Liu, L. Fei, and J. Yang, "Hyperspectral image classification based on non-parallel support vector machine," *Remote Sens.*, vol. 14, no. 10, p. 2447, May 2022.

[26] F. Tong and Y. Zhang, "Spectral–spatial and cascaded multilayer random forests for tree species classification in airborne hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4411711.

[27] J. M. Haut, S. Moreno-Álvarez, E. Moreno-Ávila, V. A. Ayma, R. Pastor-Vargas, and M. E. Paoletti, "Cloud implementation of extreme learning machine for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[28] C. Chen, Y. Ma, and G. Ren, "Hyperspectral classification using deep belief networks based on conjugate gradient update and pixel-centric spectral block features," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 4060–4069, 2020.

[29] M. Esmaeili, D. Abbasi-Moghadam, A. Sharifi, A. Tariq, and Q. Li, "ResMorCNN model: Hyperspectral images classification using residual-injection morphological features and 3DCNN layers," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 219–243, 2024.

[30] D. Wang, L. Ren, X. Sun, L. Gao, and J. Chanussot, "Nonlocal and local feature-coupled self-supervised network for hyperspectral anomaly detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 6981–6993, 2025.

[31] N. Noshiri, M. A. Beck, C. P. Bidinosti, and C. J. Henry, "A comprehensive review of 3D convolutional neural network-based classification techniques of diseased and defective crops using non-UAV-based hyperspectral images," *Smart Agricult. Technol.*, vol. 5, Oct. 2023, Art. no. 100316.

[32] R. Lei et al., "Hyperspectral remote sensing image classification using deep convolutional capsule network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 8297–8315, 2021.

[33] W.-S. Hu, H.-C. Li, Y.-J. Deng, X. Sun, Q. Du, and A. Plaza, "Lightweight tensor attention-driven ConvLSTM neural network for hyperspectral image classification," *IEEE J. Sel. Topics Signal Process.*, vol. 15, no. 3, pp. 734–745, Apr. 2021.

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[34] M. Zhu, L. Jiao, F. Liu, S. Yang, and J. Wang, "Residual spectral–spatial attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 449–462, May 2020.
[35] C. Yu, R. Han, M. Song, C. Liu, and C.-I. Chang, "FeedBack attention-based dense CNN for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5501916.
[36] W.-Y. Wang, H.-C. Li, Y.-J. Deng, L.-Y. Shao, X.-Q. Lu, and Q. Du, "Generative adversarial capsule network with ConvLSTM for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 3, pp. 523–527, Mar. 2021.
[37] C. Ding et al., "Integrating prototype learning with graph convolution network for effective active hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5504816.
[38] A. Yaghmour, S. Prasad, and M. M. Crawford, "Attention guided semisupervised generative transfer learning for hyperspectral image analysis," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 19884–19899, 2024.
[39] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2020.
[40] B. Deng, P. Duan, X. Lu, Z. Wang, and X. Kang, "Hyperspectral and SAR image classification via graph convolutional fusion network," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5538611.
[41] L. Yu et al., "DSPG: A new discriminant superpixel graph regularizer and convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5526118.
[42] A. Qin, Z. Shang, J. Tian, Y. Wang, T. Zhang, and Y. Y. Tang, "Spectral–spatial graph convolutional networks for semisupervised hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 2, pp. 241–245, Feb. 2018.
[43] Y. Ding, Y. Guo, Y. Chong, S. Pan, and J. Feng, "Global consistent graph convolutional network for hyperspectral image classification," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–16, 2021.
[44] H. Wang, Y. Cheng, C. L. P. Chen, and X. Wang, "Broad graph convolutional neural network and its application in hyperspectral image classification," *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 7, no. 2, pp. 610–616, Apr. 2023.
[45] S. Jia, S. Jiang, S. Zhang, M. Xu, and X. Jia, "Graph-in-graph convolutional network for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 1, pp. 1157–1171, Jan. 2024.
[46] Y. Dong, Q. Liu, B. Du, and L. Zhang, "Weighted feature fusion of convolutional neural network and graph attention network for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 1559–1572, 2022.
[47] E. Ouyang, B. Li, W. Hu, G. Zhang, L. Zhao, and J. Wu, "When multigranularity meets spatial–spectral attention: A hybrid transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4401118.
[48] C. Shi, S. Yue, and L. Wang, "A dual-branch multiscale transformer network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5504520.
[49] Z. Yang, X. Wu, P. Huang, F. Zhang, M. Wan, and Z. Lai, "Orthogonal autoencoder regression for image classification," *Inf. Sci.*, vol. 618, pp. 400–416, Dec. 2022.
[50] Y.-J. Deng, M.-L. Yang, H.-C. Li, C.-F. Long, K. Fang, and Q. Du, "Feature dimensionality reduction with $l_{2,p}$-norm-based robust embedding regression for classification of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5509314.
[51] D. Cheng, X. Zhang, and X. Xu, "Reweighted robust and discriminative latent subspace projection for face recognition," *Inf. Sci.*, vol. 657, Feb. 2024, Art. no. 119947.
[52] J. Bao, M. Kudo, K. Kimura, and L. Sun, "Robust embedding regression for semi-supervised learning," *Pattern Recognit.*, vol. 145, Jan. 2024, Art. no. 109894.
[53] T. G. Kolda and B. W. Bader, "Tensor decompositions and applications," *SIAM Rev. Soc. Ind. Appl. Math.*, vol. 51, no. 3, pp. 455–500, Aug. 2009.
[54] N. Han et al., "Double relaxed regression for image classification," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 30, no. 2, pp. 307–319, Feb. 2020.
[55] H. Zhang, J. Gao, J. Qian, J. Yang, C. Xu, and B. Zhang, "Linear regression problem relaxations solved by nonconvex ADMM with convergence analysis," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 2, pp. 828–838, Feb. 2024.
[56] S. J. Wright, R. D. Nowak, and M. A. T. Figueiredo, "Sparse reconstruction by separable approximation," *IEEE Trans. Signal Process.*, vol. 57, no. 7, pp. 2479–2493, Jul. 2009.
[57] L. Ren, Z. Ma, and F. Bovolo, "A novel dual-alternating direction method of multipliers for spectral unmixing," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 3, pp. 528–532, Mar. 2021.
[58] (2021). *Hyperspectral Remote Sensing Scenes*. Accessed: Jul. 7, 2023. [Online]. Available: http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes
[59] X. Li et al., "A multiscale dataset for understanding complex eco-hydrological processes in a heterogeneous oasis system," *Sci. Data*, vol. 4, no. 1, pp. 1–11, Jun. 2017.
[60] Y. Zhong, X. Hu, C. Luo, X. Wang, J. Zhao, and L. Zhang, "WHU-hi: UAV-borne hyperspectral with high spatial resolution (H^2) benchmark datasets and classifier for precise crop identification based on deep convolutional neural network with CRF," *Remote Sens. Environ.*, vol. 250, Dec. 2020, Art. no. 112012.
[61] M. Marconcini, G. Camps-Valls, and L. Bruzzone, "A composite semisupervised SVM for classification of hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 2, pp. 234–238, Apr. 2009.



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